Graph Neural Networks for Approximating Vertex Cover and Matching Problems

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Introduction: Problem Statement

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- **Vertex Cover:** A **vertex cover** $V' \subseteq V$ of an undirected graph $G = (V, E)$ is a subset of vertices such that

$$\forall \, (uv \in E) \; \Rightarrow \; \left(u \in V' \; \vee \; v \in V'\right)$$

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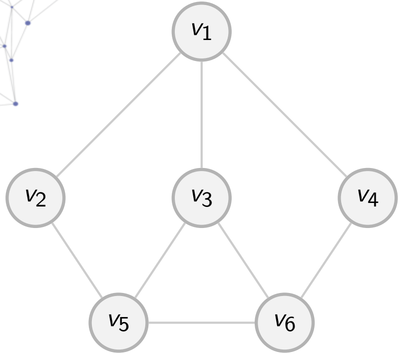
  We have to select minimum vertices covering all edges.
- **Edge dominating set:** An **edge dominating set** for a graph $G = (V, E)$ is a subset $D \subseteq E$ such that every edge $e \in E \setminus D$ is adjacent to at least one edge in D. We are finding the Minimum Edge Dominating Set .

Visualizing Minimum Vertex Cover (MVC)



MVC Goal: Select minimum nodes to touch every edge.

- Initial Graph.

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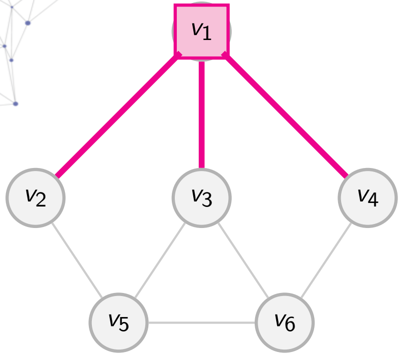

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- **Step 1:** Select v_1.

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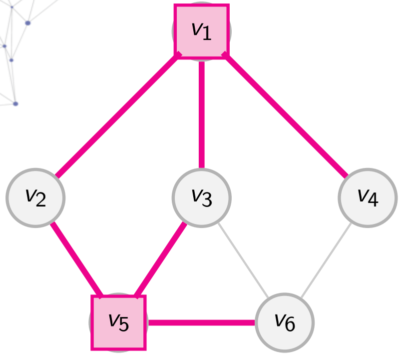

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- **Step 1:** Select v_1.
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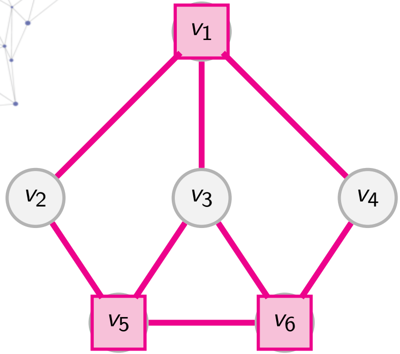

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- **Step 3:** Select v_6.

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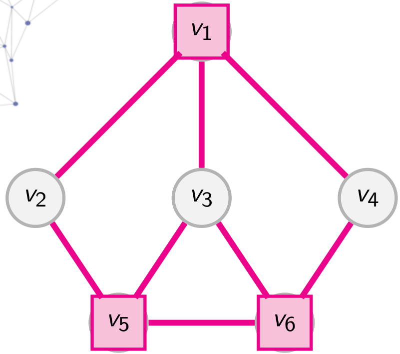

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- Initial Graph.
- **Step 1:** Select v_1.
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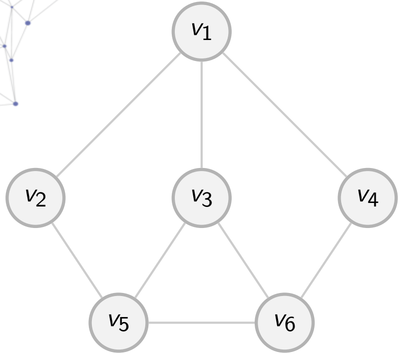
- **MVC Size = 3 Nodes**.

Visualizing Minimum Edge Dominating Set (MEDS)



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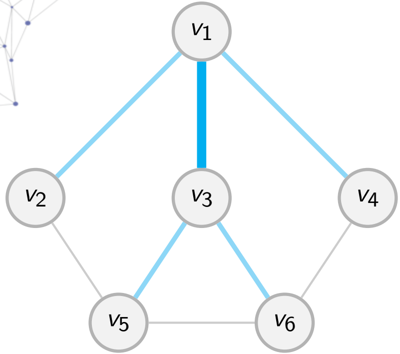
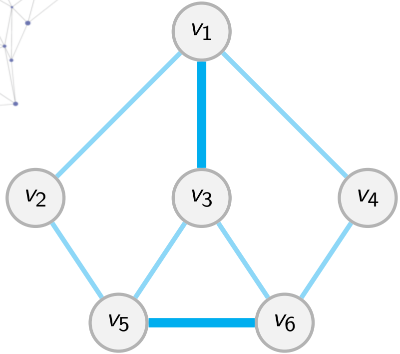


MEDS Goal: Select minimum edges to touch all edges.

- Initial Graph.
- **Step 1:** Select (v_1, v_3).

Visualizing Minimum Edge Dominating Set (MEDS)



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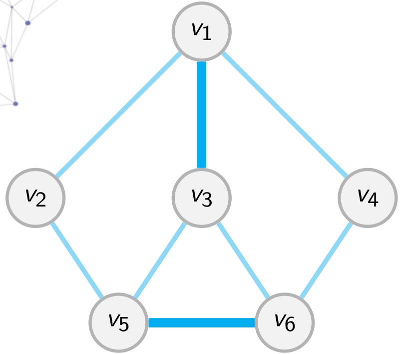

MEDS Goal: Select minimum edges to touch all edges.

- Initial Graph.
- **Step 1:** Select (v_1, v_3).
- **Step 2:** Select (v_5, v_6).
- **MEDS Size = 2 Edges**.

Graph Types Considered

- Barabási–Albert (Scale-free)

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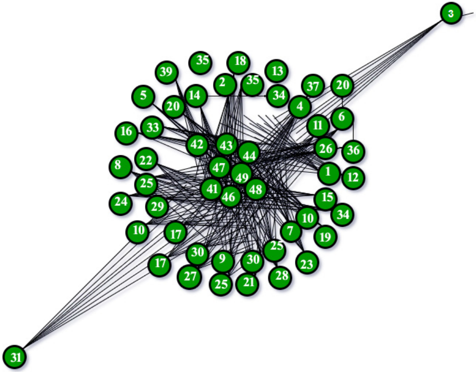
- Bipartite Graphs
- d-regular Graphs

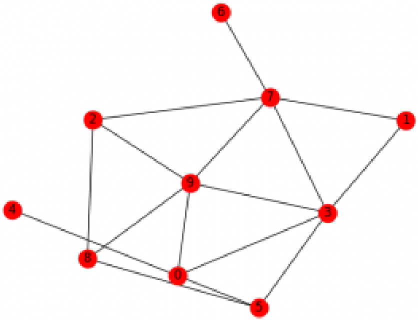
Graph Families and Their Properties

- **Barabási–Albert:**
  - Scale-free network
  - Few hubs with very high degree
  - *Example: Social networks (influencers on Instagram)*



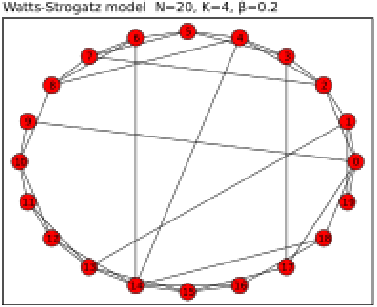
Graph Families and Their Properties

- **Erdős–Rényi:**
  - Random edge formation
  - Uniform degree distribution (on average)
  - *Example: Random communication links in a network*



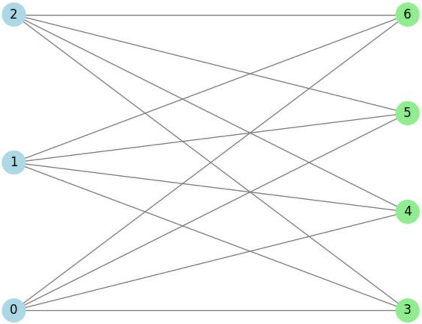
Graph Families and Their Properties

- **Watts–Strogatz:**
  - High clustering
  - Short path lengths (small-world)
  - *Example: Friend circles / small-world social networks*



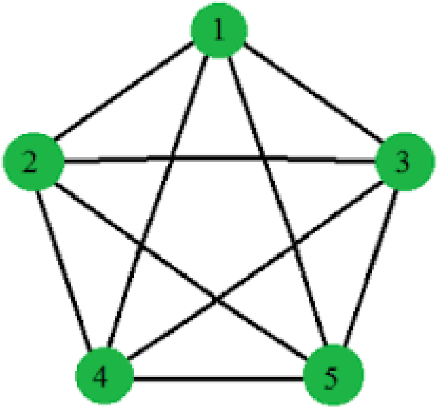
Graph Families and Their Properties

- **Bipartite Graph:**
  - Nodes split into two sets
  - Edges only across sets
  - *Example: Students–Courses enrollment network*



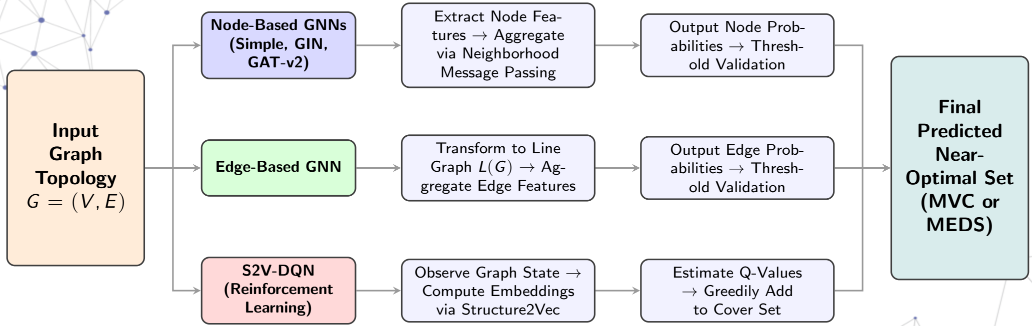
Graph Families and Their Properties

- ***d*-regular Graph:**
  - All nodes have same degree
  - Highly structured
  - *Example: Structured network topology (mesh networks)*

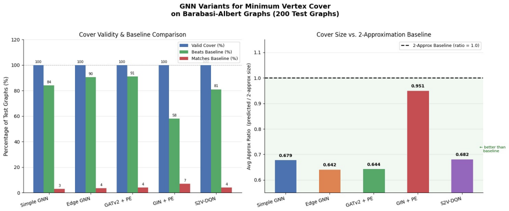


4 Regular

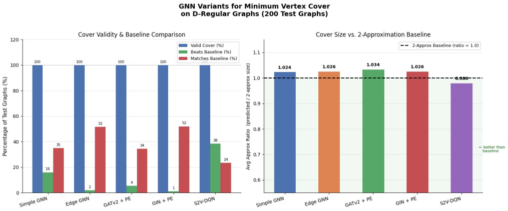
Neural Architectures Used



Minimum Vertex Cover Results: Bipartite & D-Regular Graphs



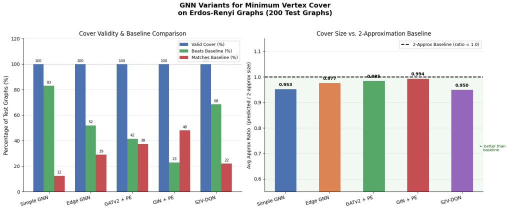
Bipartite Graphs



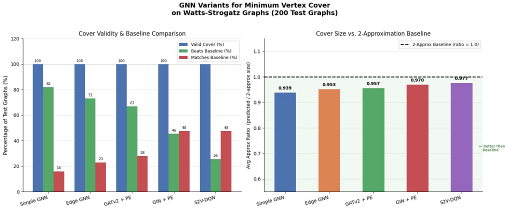
D-Regular Graphs

Figure: Performance comparison of GNN variants vs. 2-Approximation baseline.

Minimum Vertex Cover Results: Erdős-Rényi & Watts-Strogatz



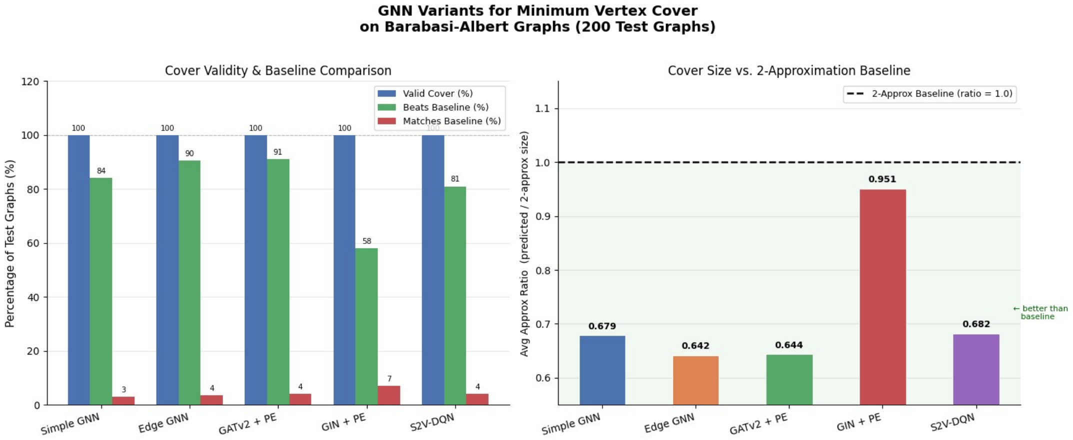
Erdős-Rényi Graphs



Watts-Strogatz Graphs

Figure: Performance comparison of GNN variants vs. 2-Approximation baseline.

Minimum Vertex Cover Results: Barabási-Albert Graphs



Comparison with State-of-the-Art

Graph Family	Existing Work	Our Approach
Barabási-Albert *(Scale-Free)*	Dai et al. (S2V-DQN) achieved approximation ratios of 1.01–1.05 compared to exact CPLEX solutions.	**Edge-Based GNN** significantly outperforms the 2-approx baseline, achieving an average ratio of **0.642**.
Bipartite	Dai et al. formulated the Set Covering Problem, achieving ratios ranging from 1.001–1.07.	**Simple GNN** leverages the structure exceptionally well, reaching a highly optimal ratio of **0.629**.

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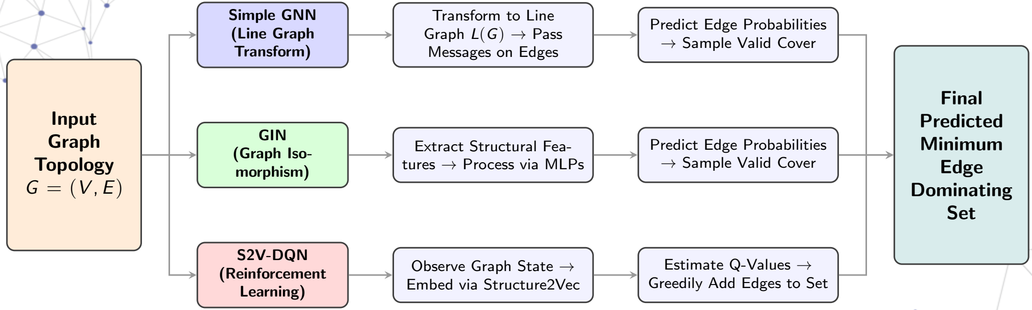
Graph Family	Existing Work	Our Approach
Watts-Strogatz *(Small-World)*	Yau et al. (OptGNN) evaluated MVC on these networks, achieving an empirical ratio of $\sim$1.031.	**Simple GNN** achieves an average ratio of **0.939** vs the baseline, demonstrating robust heuristics.
D-Regular & Erdős-Rényi	Martin et al. (Physics-Inspired GNNs) achieved estimated approximation ratios of $\sim$0.88 to 0.92.	**S2V-DQN** maintains highly competitive performance, achieving sub-1.0 ratios (**0.980** and **0.950**).

Approximation of Minimum EDS: Neural Architectures Deployed



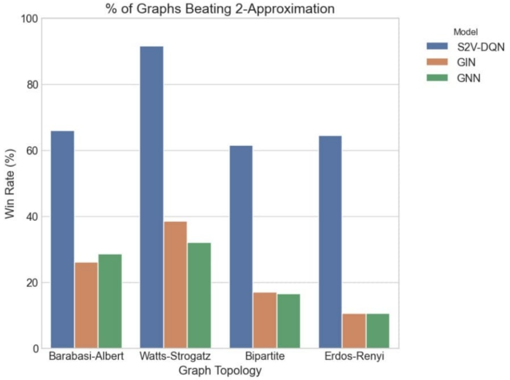
The Benchmark: 2-Approximation Algorithm

Criteria	Details
Lack of Prior Work	No existing deep learning literature for the MEDS problem; benchmarking strictly against classical algorithms.
Mathematical Standard	Exclusively benchmarked against the classical **2-Approximation** algorithm.
Mechanism	Finds an arbitrary maximal matching (fast, but mathematically rigid).
Objective	Demonstrate GNNs can autonomously discover more optimal structural patterns than this standard.

Key Experimental Results (vs. 2-Approximation)

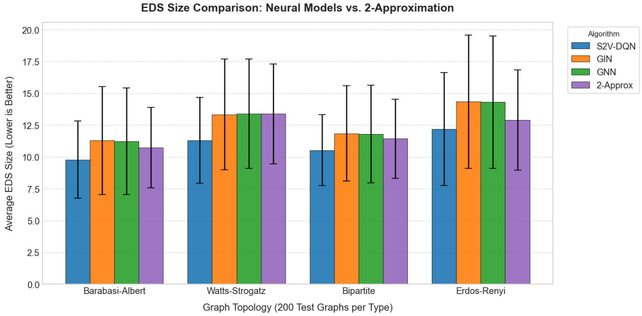
Graph Topology	**Performance vs. 2-Approximation**
Bipartite	Message-passing GNN yielded our tightest average approximation ratio of **1.124**.
Watts-Strogatz	Neural models found a strictly smaller, more optimal cover on **32.0%** of test graphs.
Barabási-Albert	Neural models outperformed the 2-Approximation baseline on **28.5%** of test graphs.
Overall Trend	S2V-DQN and Simple GNNs consistently find superior, smaller covers than the local maximal matchings found by the baseline.

Performance Overview: Dominance Metrics (Win Rates)



- **Key Takeaway:** S2V-DQN achieves a massive 91.5% win rate against the 2-Approximation algorithm on Watts-Strogatz graphs.

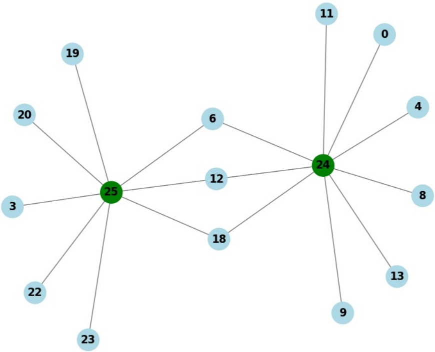
Performance Overview: Average EDS Size Comparison



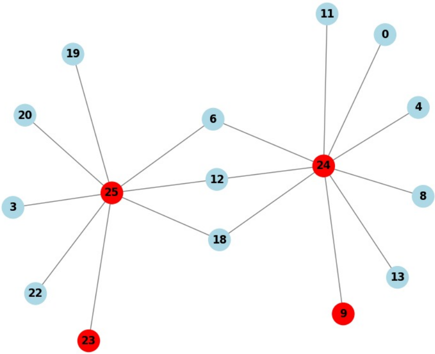
- S2V-DQN maintains the lowest (most optimal) average EDS size across all tested topologies.
- Standard node-based GNN/GIN architectures struggle severely with random Erdős-Rényi graphs.

Visualizing (Vertex Cover): GNN vs. 2-Approximation

Model: Edge-Based GNN | Graph Size: 16 | VC Size: 2



2-Approximation Algorithm | Graph Size: 16 | VC Size: 4



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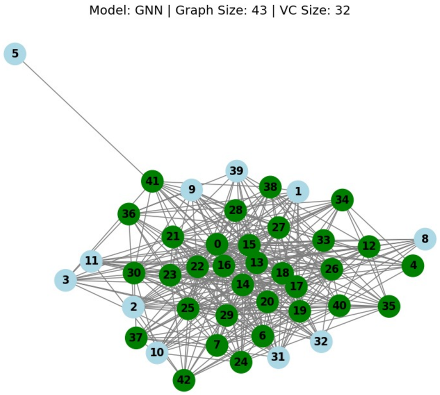
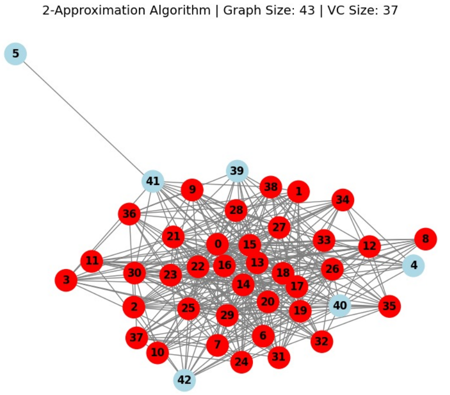
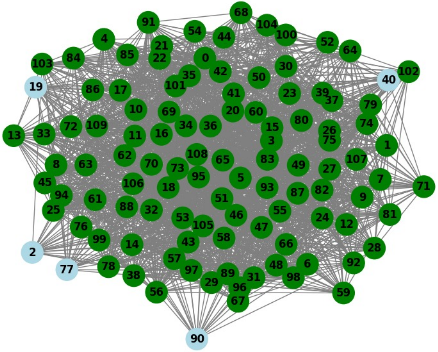


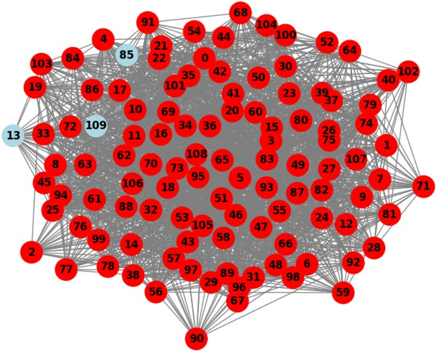


Visualizing (Vertex Cover): GNN vs. 2-Approximation

Model: GATv2 | Graph Size: 110 | VC Size: 105



2-Approximation Algorithm | Graph Size: 110 | VC Size: 107



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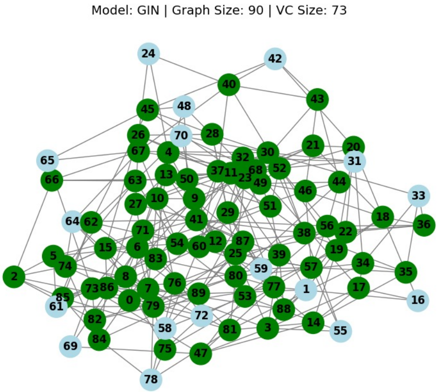

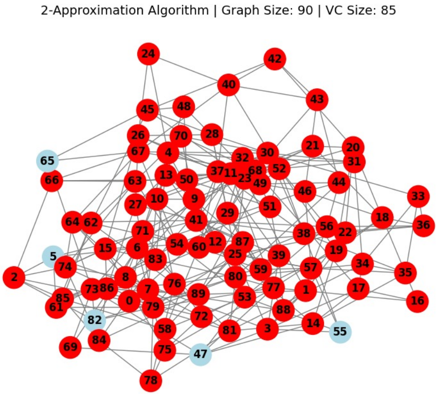

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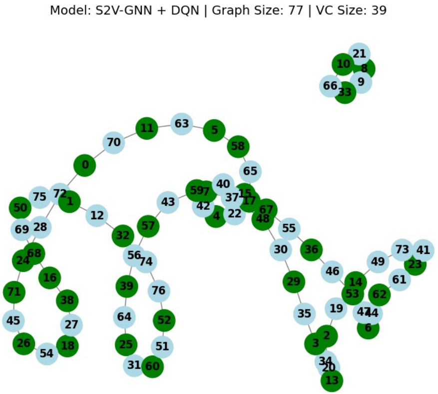
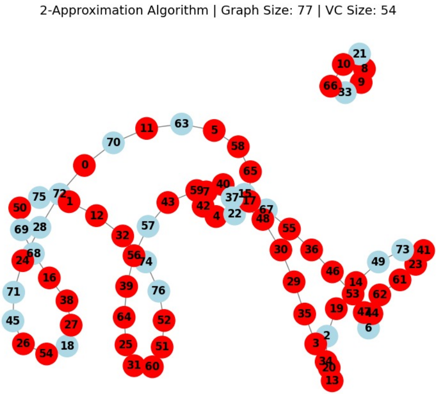
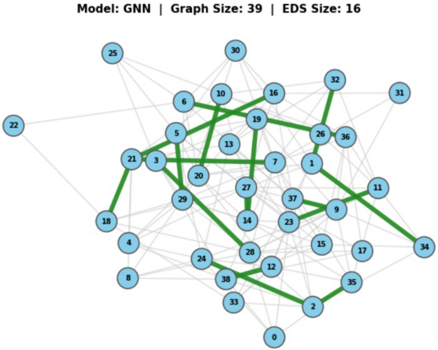
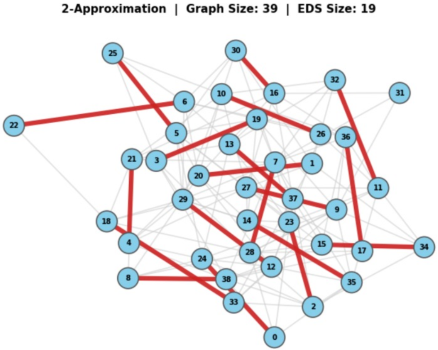




Visualizing (EDS): S2V-DQN vs. 2-Approximation





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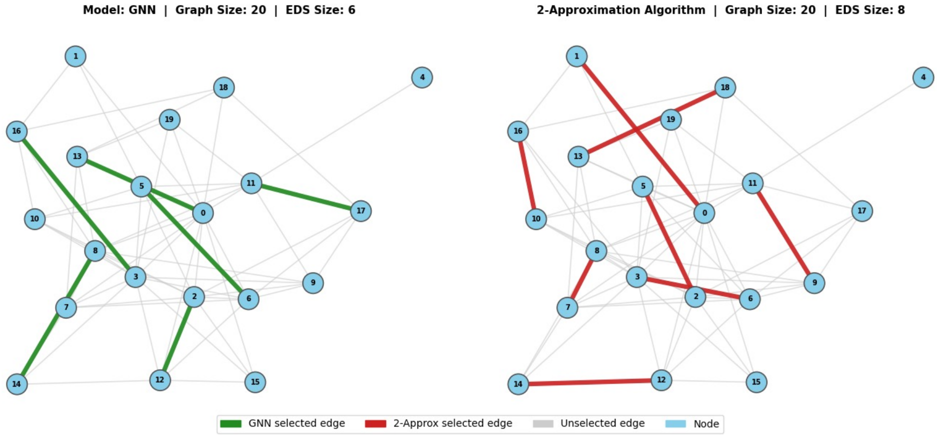

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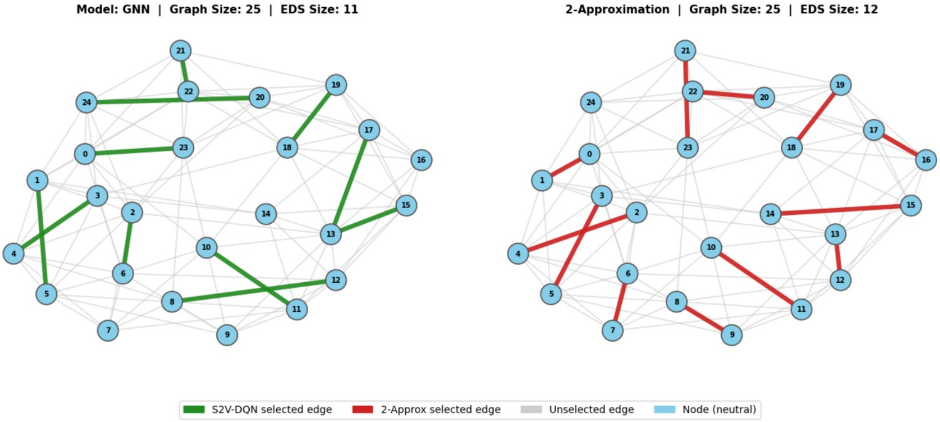

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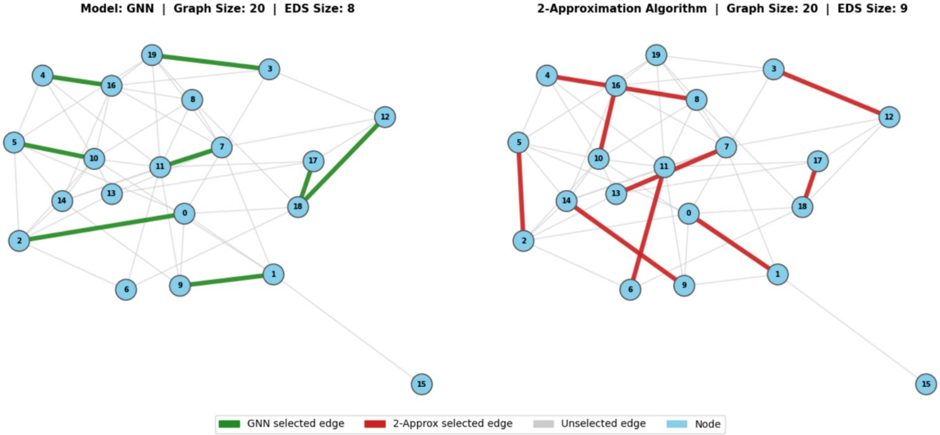

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- **New Standard:** Neural models consistently beat the 2-Approximation baseline.
- **Topology matters:** Simple GNNs dominate Bipartite; S2V-DQN dominates Small-World.
- **Future Path:** With a 91.5% win rate, Reinforcement Learning is the leading approach.

References I

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Thank You